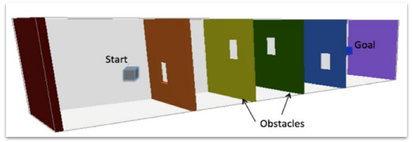
A Literature Review on Motion Planning and Intention Prediction Methods

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Introduction and Background

Motion Planning (MP) is the process of algorithmically computing a trajectory for an agent to get from a start point to a goal point.



Intention Prediction (IP) is the process of using different probabilistic frameworks to anticipate the actions of either surrounding agents or the ego agent

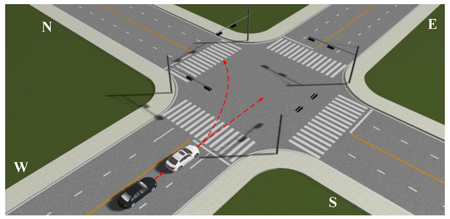


Figure 1. Top: Example of a motion planning problem. Bottom: Example of intention predictions for autonomous vehicles. (Source: MDPI)

Motivation

1. Are there benefits to integrating IP + MP?
2. Are there standardized terms for these methods?
3. How do we integrate IP + MP?

Initially, we aimed to investigate the integration by implementing some methods that integrated IP + MP. In doing this however, we ran into two issues: volume of methods of integration and standardization of terms.

As such, we conducted a literature review of just over 70 papers to investigate these issues as well as formalize our findings and answer our questions above.

We focused on papers that met the following criteria:
- Does it integrate MP and IP in the core method
- Balanced view of both MP and IP
- Modular approaches to the integration
- Handle uncertainty

Sequential and Modular Methods

Overview:
- IP and MP operate as completely separate modules
- Prediction output feeds directly into the planner as a fixed input

1. Deterministic Tracking
- Uses kinematic models (e.g., constant velocity/acceleration) or Kalman Filters
- Projects current state forward in time
- Produces a single predicted trajectory per agent.

2. Probabilistic Modules
- Models the future as a distribution of possible states
- Uses Gaussian Mixture Models, dynamic occupancy grids, etc.
- Outputs a probability map or set of weighted trajectories
- Gives the planner a richer picture of likely futures

Limitations:
- **Error Propagation**: Errors and information loss from the prediction modules get amplified and/or accumulate, eventually contributing to suboptimal motion.
- **Robot Freezing:** Planners tend to freeze because probabilistic predictions expand over time and may not allow any feasible paths.
- **Solution Attempts:** Minor improvements are made by implementing short-horizon replanning or applying more conservative risk thresholds to prevent complete immobilization.

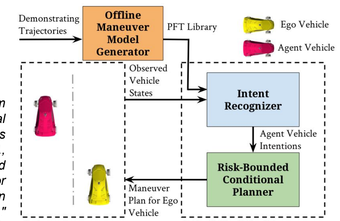


Figure 1. System Overview of a sequential model. This figure is from #54, Huang et. al., "Online risk-bounded motion planning for autonomous vehicles in dynamic environments." ICAPS 2019.

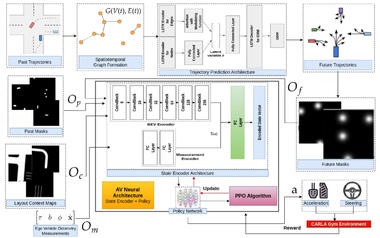
Joint and Integrative Methods

Primary Objective: To move beyond serial processing by using Integrated Prediction and Planning where modules share representations or are optimized simultaneously.

1. Core Methodologies
- **Markov Decision Processes (MDPs):** Model the history of the system via the current state. States depend only on that state and the chosen action.
- **Partially Observable MDPs (POMDPs):** Applied when the agent receives partial and noisy observations.
- **Inverse Reinforcement Learning (IRL):** Infers reward function from human demonstrations.
- **Mixed Observability MDP (MOMDP):** Robot solves for a model incorporating unobservable beliefs over human goals

2. Integrated Collaborative Features
- **Trust in the Robot:** Temporally evolving metric, often modeled via dynamic Bayesian network
- **Human Expertise:** Parameter estimating proficiency through analysis of demonstrated trajectories

3. Real-world Applications
- **Autonomous Vehicle Pedestrian Interaction:** MOMDPs to maintain a distribution over a pedestrian's intent.
- **Factory Assembly Line Collaboration:** POMDPs to infer the human's intended assembly sequence from object selections.
- **Trust-Aware Assistive Robot Assistance:** Assistive robots use POMDP formulations to track a user's trust as a temporally evolving metric, adapting the robot's motion policy to ensure team success.



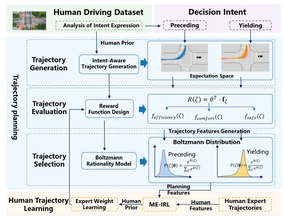


Figure 2a (left): Example of POMDP obsesrvations to robot decisions. GP3Net encodes several types of input as observations. This figure is from #4, Chowdhury et. al., "Graph-based Prediction and Planning Policy Network (GP3Net) for scalable self-driving in dynamic environments using Deep Reinforcement Learning" AAAI 2024.

Figure 2b (right): An Inverse RL framework for trajectory prediction and planning. This figure is from #37, Liu et. al., "Human-Like Implicit Intention Expression for Autonomous Driving Motion Planning Based on Learning Human Intenion Priors" IEEE T-IV 2025.

End to End (E2E) Methods

Overview:
- No separate IP or MP modules - handled implicitly by deep neural networks
- Input: raw or lightly processed sensor data (camera, LiDAR, radar, rasterized maps)
- Output: trajectories

1. RNNs & LSTMs
- Designed for sequential data - capture temporal trends over time
- Encode historical trajectories of agents and ego-vehicle
- Model velocity and movement patterns

2. Graph Convolutional Networks (GCNs)
- Agents = nodes (encoding position, velocity, type, bounds)
- Interactions = edges
- Enables relational and spatial reasoning between agents

3. Attention Mechanisms
- Prioritizes relevant actors (e.g., nearby pedestrian) over background noise
- Improves safety and computational efficiency
- Can substitute for GCNs or integrate into larger architectures

4. Hierarchical Integration
- Hybrid E2E structure with two network levels
- Upper layer: predicts high-level intents (e.g., turn left, yield)
- Lower layer: generates continuous control outputs (steering angle, acceleration)).

Conclusions & Future Work

Results of these integrated methods showed:
- decreased collision rates
- farther distances between robots working collaboratively with humans.

A large mix of different terms were used in order to refer to the same underlying process of intention prediction. Standardizing these terms will greatly increase the ease of discovery within this space. Furthermore, the promising results show that developing these methods is critical, and potential work towards a standard algorithm for Autonomous Driving and HRC could be work for the future.

Citations and Acknowledgements

All works can be found at https://bit.ly/ipmpsurvey.

We would like to thank the UIUC CS STARS program as well as the CRA's UR2PhD program for providing us funding and support. We would also like to thank our research mentors Nancy Amato and Ananya Yammamuru for their help and guidance through this project.

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